

DUMAGUETE, Philippines

WMO: 986420

Lat: 9.334N

Lon: 123.300E

Elev: 5

StdP: 101.27

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.0	23.5	20.9	15.5	24.9	21.5	16.1	25.2	9.8	26.4	8.1	26.8	0.6	320	0.347

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	6.3	33.3	27.2	32.8	27.2	32.2	27.1	28.2	31.9	27.8	31.6	27.6	31.4	3.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	22.9	31.0	26.7	22.3	30.7	26.4	21.9	30.4	90.8	32.1	89.0	31.9	87.5	31.6	31.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.2	6.0	5.3	19.7	35.7	2.5	2.2	17.9	37.3	16.5	38.6	15.1	39.9	13.3	41.5
			18.9	29.3	2.3	0.9	17.3	29.9	15.9	30.4	14.7	30.9	13.0	31.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.8	26.8	26.9	27.4	28.2	28.7	28.3	27.9	28.1	28.0	27.8	27.8	27.5
	DBStd	1.05	0.90	0.89	0.84	0.86	0.93	0.98	0.93	1.06	0.90	0.90	0.86	0.85
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6492	522	473	541	547	579	549	553	560	538	552	534	543
	CDD18.3	3450	263	240	282	297	321	299	295	302	288	293	284	285
	CDH23.3	37957	2487	2344	3034	3566	3952	3488	3233	3420	3139	3162	3133	2999
	CDH26.7	12848	492	534	905	1351	1653	1370	1156	1332	1131	1087	1018	819
(14) Wind	WSAvg	2.1	2.6	2.5	2.3	2.1	1.9	1.7	1.7	1.9	1.7	1.7	2.0	2.4
(15) Precipitation	PrecAvg	1215	87	67	65	38	84	103	142	113	131	157	155	98
	PrecMax	1882	285	261	212	138	185	218	308	261	521	354	282	183
	PrecMin	672	5	13	3	4	9	34	12	49	42	44	71	35
	PrecStd	316	77	63	69	34	52	51	83	54	115	103	72	43
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.9	31.0	31.7	32.7	33.9	34.0	33.7	34.1	33.5	32.9	32.1	31.7
		MCWB	26.4	26.3	26.6	27.4	27.4	27.1	27.2	27.1	27.3	27.2	26.9	27.0
	2%	DB	30.1	30.2	31.1	32.0	32.8	33.0	32.6	33.1	32.7	32.0	31.4	31.1
		MCWB	26.2	26.0	26.3	27.0	27.4	27.0	27.0	27.1	27.1	27.0	26.9	26.8
	5%	DB	29.6	29.9	30.6	31.4	32.2	32.2	31.9	32.2	32.0	31.4	31.0	30.4
		MCWB	26.0	25.8	26.1	26.8	27.3	27.2	26.9	27.0	27.0	26.9	26.8	26.6
	10%	DB	29.0	29.3	30.1	31.0	31.7	31.5	31.1	31.6	31.2	30.9	30.5	29.9
		MCWB	25.7	25.6	25.9	26.6	27.2	27.0	26.8	26.8	26.8	26.8	26.6	26.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.1	27.0	27.3	28.1	28.7	28.4	28.1	28.5	28.3	28.1	27.7	27.7
		MCDB	29.8	30.1	30.8	31.9	32.1	32.3	32.0	32.5	32.0	31.6	30.9	30.6
	2%	WB	26.6	26.5	26.8	27.5	28.1	27.8	27.5	27.7	27.7	27.5	27.2	27.1
		MCDB	29.4	29.6	30.2	31.2	31.9	31.9	31.5	31.9	31.7	31.1	30.5	30.2
	5%	WB	26.2	26.1	26.5	27.1	27.7	27.5	27.1	27.2	27.2	27.1	27.0	26.8
		MCDB	28.9	29.2	29.8	30.8	31.5	31.4	31.1	31.5	31.2	30.8	30.4	29.8
	10%	WB	25.8	25.8	26.2	26.8	27.3	27.1	26.8	26.9	26.9	26.8	26.7	26.4
		MCDB	28.5	28.8	29.4	30.3	31.1	31.0	30.6	31.1	30.7	30.3	29.9	29.3
(35) Mean Daily Temperature Range	5% DB	MDBR	4.3	4.7	5.3	5.6	6.3	6.6	6.6	7.1	6.8	6.3	5.5	4.8
		MCDBR	5.1	5.6	6.1	6.1	7.0	7.5	7.8	8.3	7.9	7.2	6.1	5.6
		MCWBR	2.7	2.9	3.2	3.0	3.6	3.7	3.8	4.1	4.0	3.6	3.1	2.9
	5% WB	MCDBR	4.7	5.1	5.5	5.8	6.6	7.1	7.3	7.8	7.4	6.8	5.9	5.3
		MCWBR	2.7	2.9	3.0	3.1	3.6	3.8	3.8	4.1	4.0	3.6	3.1	2.9
(40) Clear-Sky Solar Irradiance	taub		0.423	0.418	0.421	0.432	0.431	0.421	0.421	0.440	0.436	0.448	0.430	0.426
	taud		2.455	2.456	2.451	2.419	2.433	2.475	2.477	2.403	2.408	2.372	2.437	2.447
	Ebn at Noon		888	904	901	879	863	861	865	860	874	863	875	877
	Edh at Noon		113	116	118	120	116	110	110	121	121	124	114	112
(44) All-Sky Solar Radiation	RadAvg		4.31	5.09	5.82	6.20	5.60	5.08	4.84	5.00	4.90	4.84	4.73	4.36
	RadStd		0.46	0.50	0.46	0.25	0.38	0.27	0.31	0.37	0.45	0.33	0.36	0.36

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.22	N/A	N/A	+0.61	+0.47	+0.39	N/A	N/A	N/A
(47) Regional (7 neighbors)		+0.15	N/A	N/A	+0.40	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page